

University of Illinois at Chicago: CHE-494 Biopharmaceuticals Process Development and Manufacturing (BPDM)

Instructors: Dr. Nandkishor Nere and Professor Meenesh Singh

Spring 2026, January 13th - May 5th

Tuesdays at 5:00 pm to 7:45 pm (1:15 min x 2; 15 min of break)

Course Schedule

Week	Module	Topics to be covered
Week 1	Module 1	• Introduction to BPDM (NK Nere and Meenesh) • Expectations, projects-grad & undergrad, discipline specific • Discovery, Process, Drug delivery – Course Focus on CMC and intro to CMC • Power of why?
Week 2	Module 1	Bio/Analytical Chemistry Tools and Technologies
Week 3	Module 2	API-Solid Form Screening and Selection
Week 4	Module 2	Essentials of API Process Development • Fundamentals of chemical reaction engineering (CRE) • Applications of CRE in pharmaceutical industry
Week 5	Module 2	Essentials of API Process Development • Crystallization process development
Week 6	Module 2	Essentials of API Process Development • Extraction, Distillation, Filtration, Washing and Drying • Process Scale-up
Week 7	Module 2	Essentials of Biologics Process Development • Upstream process development • Downstream process development
Week 8	Module 3	Essentials of Small Molecule Drug Product Process Development • Formulation Screening and Selection Project Kick-off Discussion
Week 9	Module 3	Essentials of Small Molecule Drug Product Development Part I: • Blending • Roller compaction • Wet Granulation • Film coating Part II: • Spray Drying • Hot Melt Extrusion • Tableting and Encapsulation

Week 10	Module 3	Essentials of Biologics Drug Product Development • Formulation, overall process and lyophilization Alternate: Process Scale-up and Project Touch Point Discussion (after combining with week 11)
Spring Break		
Week 11	Module 3	Essentials of Biologics Drug Product Development • Freeze-Thawing • Pooling, Reconstitution, Filling
Week 12	Module 4	Advanced Topics: • Applications of Predictive Modeling
Week 13	Module 4	Advanced Topics: • Continuous Processing • Emerging Technologies-Electrochemistry, Photochemistry, Bio-catalysis, Allied separations, Green chemistry
Week 14	Module 5	• cGMP and key Considerations for the Safety and Cleaning in Manufacturing • Project Presentations -I
Week 15	Module 5	• Project Presentations -II • Feedback

